CFD Post-Processor for NPSH Analysis

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1. Purpose.

The document describes a guideline for post-processing the results obtained from a CFD simulation activity with a cavitation model activated.

2. Scope.

This procedure is applicable to any centrifugal pumps. It is used specifically for the first stage of the pump which is subject to the cavitation phenomenon.

The procedure is applicable for the Nova Milanese and La Spezia plants.

3. Abbreviations.

1ST		*First Stage model*
CFD		*Computational fluid Dynamics*
Eta	*[%]*	*Efficiency*
H	*[m]*	*Head*
NPSH	*[m]*	*Net Positive Suction Head*
NPSH3%	*[m]*	*Net Positive Suction Head 3% (H decreases by 3% due to cavitation)*
NPSHA	*[m]*	*(Net Positive Suction Head Available)*
NPSHi	*[m]*	*Net Positive Suction Head inception*
Nq or Ns		*Specific speed of the pump*
Q	*[m^3/h]*	*Flowrate*
SSS		*Suction Specific Speed*
VOF		*Volume of Fraction*

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4. General procedure: description.

The guideline provides specific instructions for the post-processing of the results obtained from a CFD simulation with a cavitation model.

It explain how to manage the results of the simulations prepared using the SOP "CFD Pre-Processor for NPSH analysis"; it is important to remember that the calculation setting of the previous SOP were referred to a simplified first-stage model called "1ST",with a steady-state analysis and the setting of the multiphase water-vapour fluid model (VOF method).

The main target is to estimate the NPSH3% and NPSHi with sufficient accuracy, as close as possible to the experimental results.

The adopted methodologies include the use of the "Head Drop Curve" and the visual investigation of the cavitation bubbles by 3D visualization and CFD post-processing graphical tools.

A specific discussion is presented below, with the aim to provide the user with the necessary information and criteria adopted for the estimation of the entity of cavitation using a specific purpose tools of Post-Processing “CFX-Post” inside the environment ANSYS.

5.1. NPSH3% Analysis

Before describing the standard method adopted for the estimation and evaluation of NPSH3% is necessary to define two variables of interest and their formulation:

- NPSH (Net Positive Suction Head)

$$NPSH = \frac{Pressure_{tot_{inlet_{suction}}}[Pa] - Pressure\ Vapour[Pa]}{density\ water\left[\frac{kg}{m3}\right] * gravitational\ acceletation[\frac{m}{s2}]}$$

$$Water\ pressure\ Vapour\ at\ 25°C = 3167\ [Pa]$$

$$density\ water\ at\ 25°C = 997\left[\frac{kg}{m3}\right]$$

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$$gravitational\ acceletation = 9.81\ [\frac{m}{s2}]$$

- NPSH3% (Net Positive Suction Head 3% (first stage Head fall down by 3% due to cavitation))

- NPSHA (Net Positive Suction Head Available)

- SSS (Suction Specific Speed)

$$SSS = speed\ shaft\ [rpm] * \frac{(Mass\ Flow\ Impeller\ [\frac{m3}{s}]|@bep)\text{^}0.5}{(NPSH\ [m])\text{^}0.75}$$

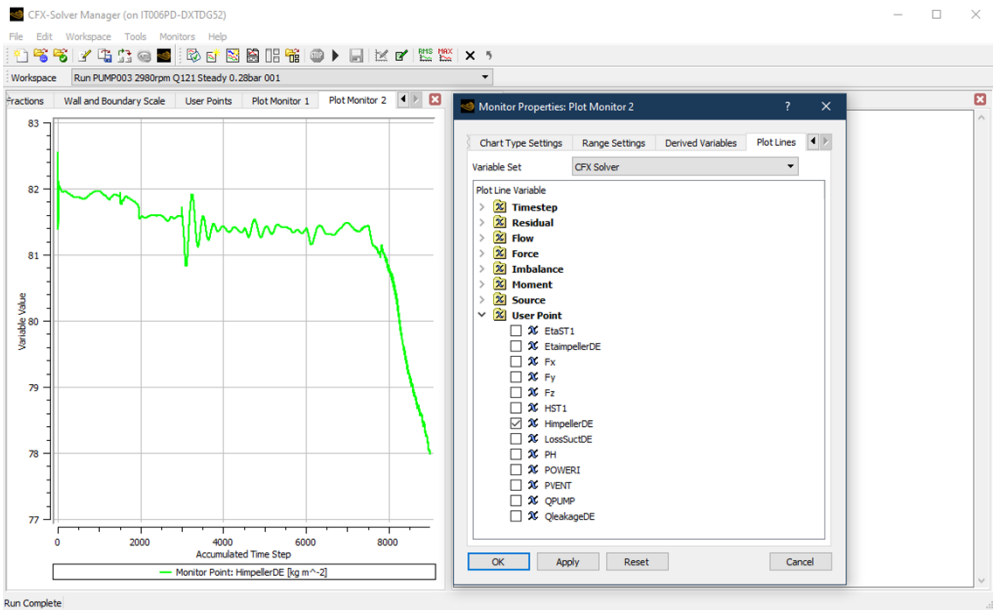
5.2. Numerical Methodology: Head Drop Curve

The criterion adopted to create the "Head Drop Curve" consists in progressively reducing the Total Pressure at the suction's inlet and measuring the Head for each of drop point respectively.

To reduce calculation times, it is necessary to define a limited number of drop points.

Measurement of the Head for each drop point

Evaluate with an average the convergence value of each drop point, considering that each of them requires 1000 to 1500 iterations and start with the previous simulation according to the sequence described in Table 1.



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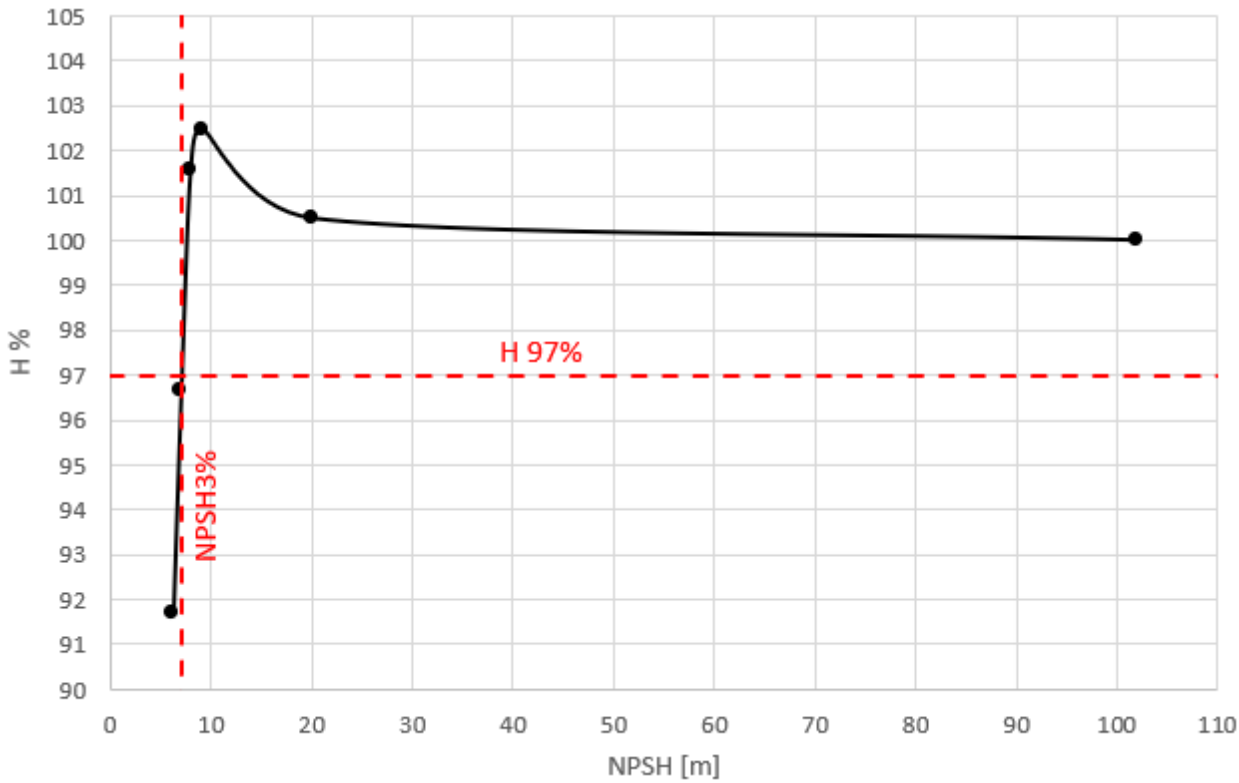
Therefore, complete the column relating to the Head:

Pasp [Pa]	NPSH [m]	SSS	H [m]	H [%]
1000000	101.9	30	111.26	100
200000	20.1	100	111.80	100.483
93000	9.2	180	114.00	102.461
81000	8.0	200	113.00	101.562
72000	7.0	220	107.53	96.6491
64400	6.3	240	102.00	91.6753

graphic drawing of the “Head Drop Curve”



Identification of the NPSH3%

As per definition the value of NPSH3% must be sought in the curve as that value of NPSH where the Head is reduced by 3% compared to the condition of non-cavitating Head.

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Other rules to adopt to build an accurate curve and to speed up the simulation activity are:

- since all simulations are stationary a verification of the convergence of each calculation is necessary to identify the appropriate value of Head.

- if the drop points were not sufficient to define the curve it is necessary to prepare other drop points to be interposed in the sequence.

- if the simulation activity is being monitored live and the curve reaches H97% in advance it is not necessary to simulate the drop points set downstream.

Usually, it is required to realize a NPSH3% curve for different mass flow values, in this case it is necessary to set up the simulations, similarly, keeping in mind in advance that the SSS value far from the BEP is generally lower than the SSS of BEP, especially for high mass flows.

5.3. Visual Methodology: cavitation bubbles

The criterion adopted is based on the use of the concept of volume fraction VF, that is, the volume fraction of water and vapour that share and fill the individual cells of the mesh.

For the graphic visualization of the bubbles follow the following steps:

1) Multiple visualization of the vane of impeller

Since the simulated pump model 1ST consists of only a single vane of impeller it is necessary to create a multiple view in order to display the complete impeller.

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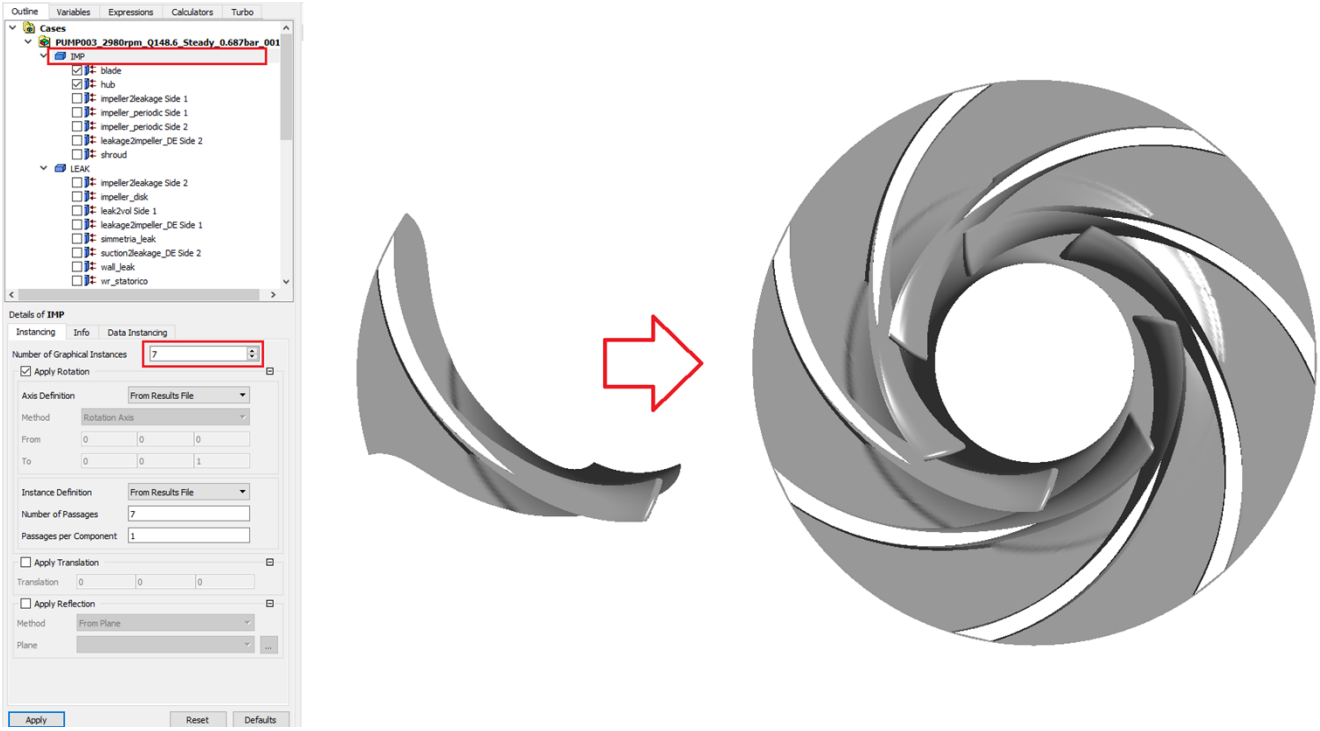
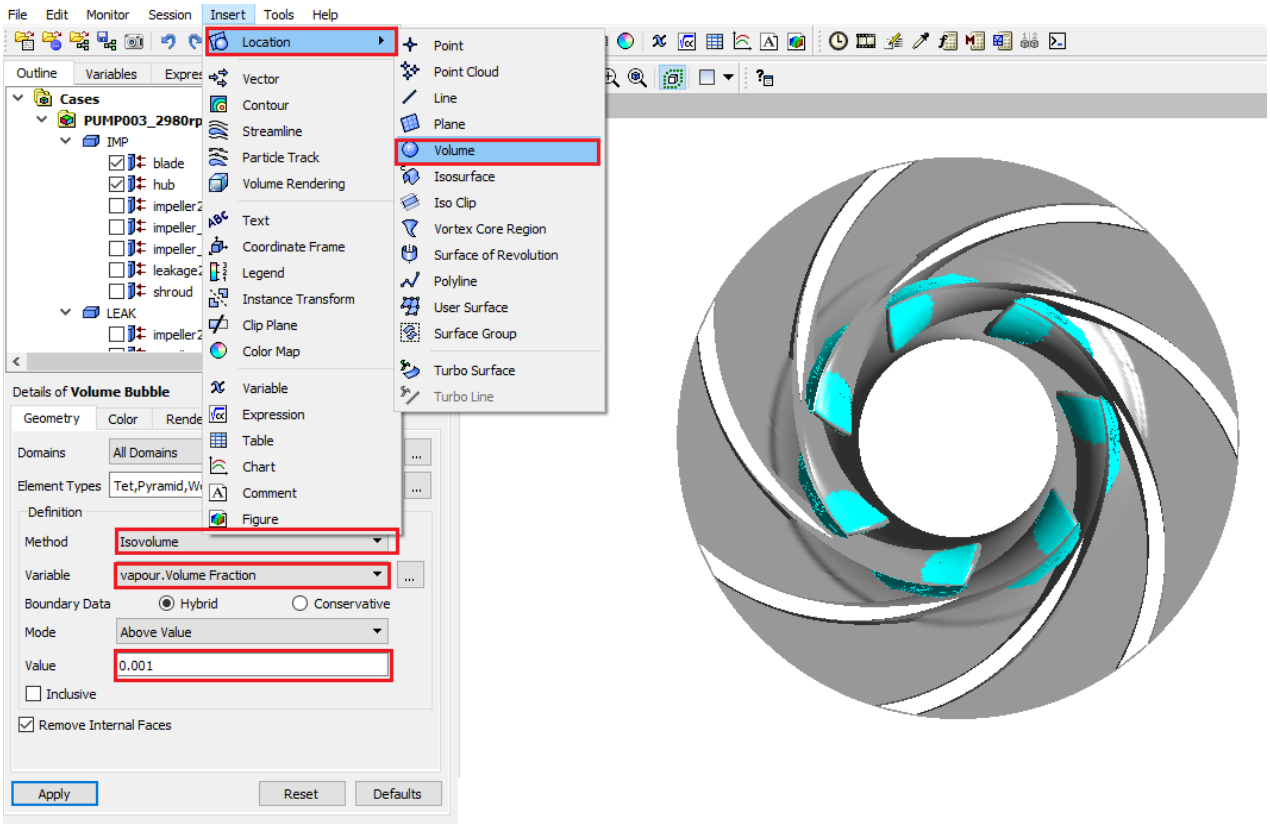


2) Creation of the “Isovolume”

To create the isovolume follow the setting as shown in the figure:



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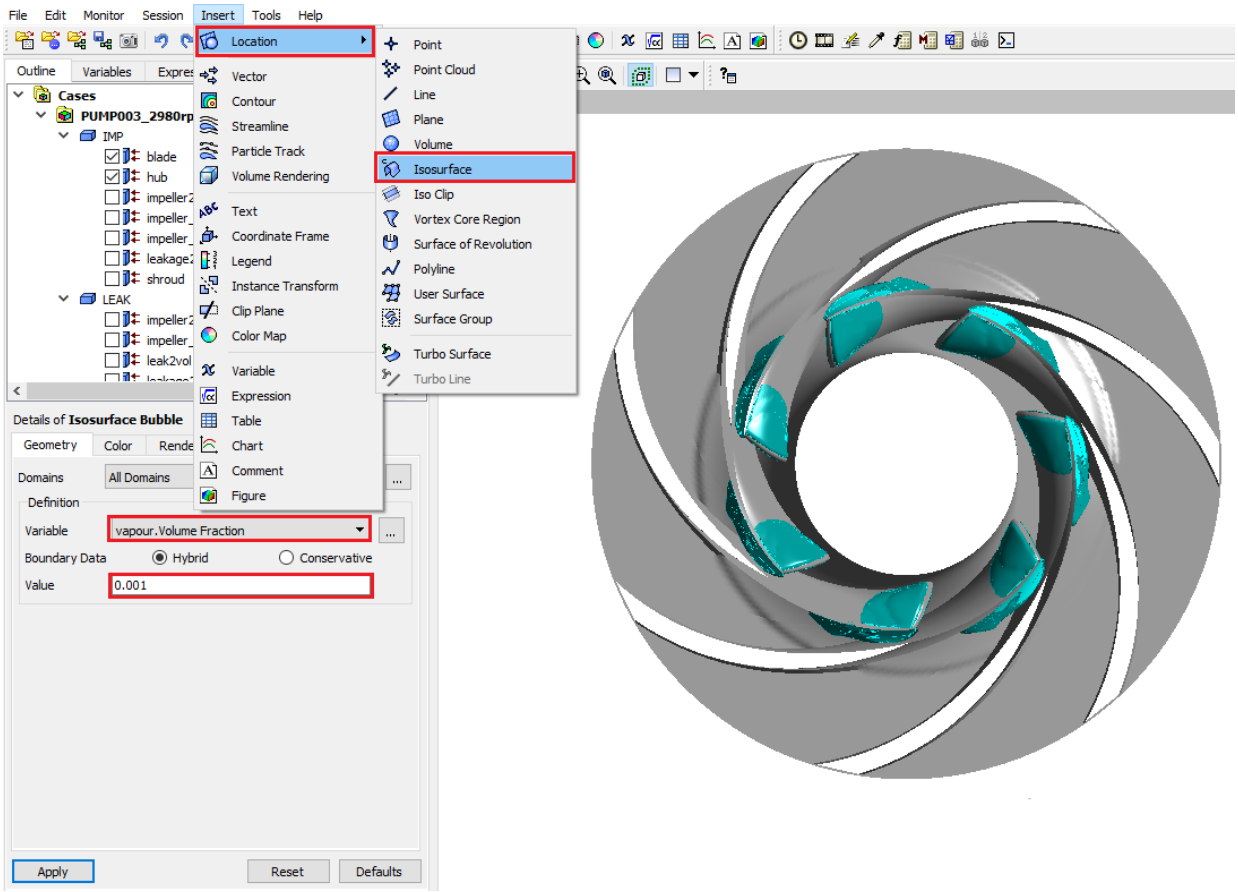
3) Creation of the “Isosurface”

To create the isosurface follow the setting as shown in the figure:



4) Visual analysis of cavitation bubbles

To determine the NPSH3% using this methodology a visual analysis of the cavitation bubbles is necessary.

By analysing the individual results of the drop points the value of NPSH3% is determined for that value of Total pressure in suction that causes the occlusion of the vane of impeller, that is, it occurs when the set of bubbles identified by the isovolume touches the blade of the next vane.

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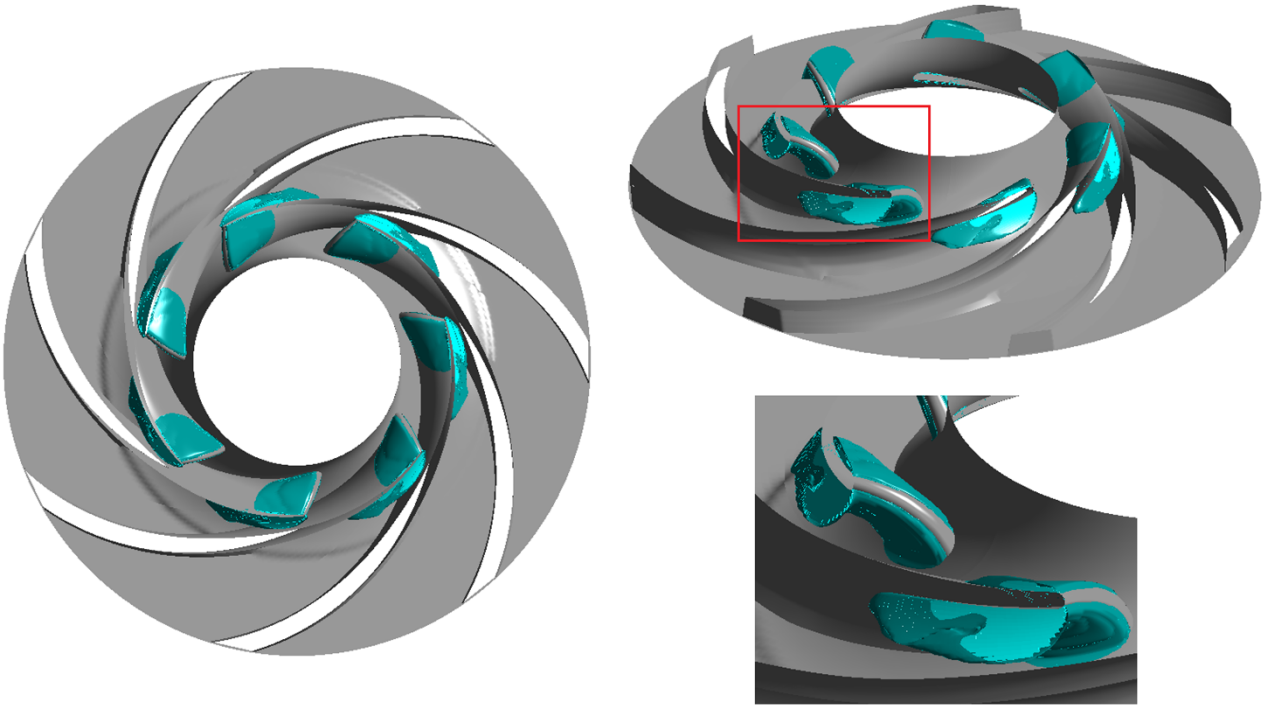

Once the two methods have been performed, the identification of NPSH3% focuses on the more conservative method.

5. NPSHi Analysis

The inception NPSHi is defined as that NPSH for which the first formation of cavitation bubbles occurs, that is, when they assume an extension of length equal to 10 [mm].

The identification of the NPSHi can be obtained by following the same visual methodology described for the NPSH3%.

Below a figure taken from theory [1] shows on position 1 where the formation of cavitation bubbles begins and their extension occurs.

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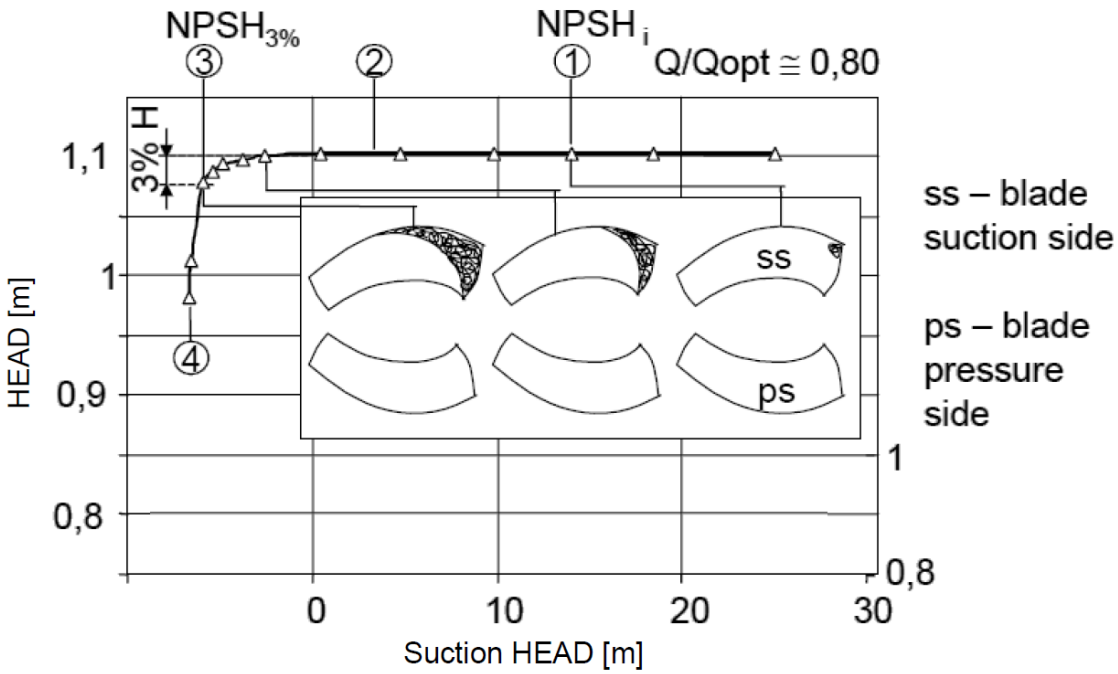

6. Setting files for batch mode

During the redesign and optimization phase of a particular hydraulic pump component, there is usually a tendency to repeat the preparation phases of the pump model and its setting.
Since the investigation concerns the single component, it is possible to prepare a series of modifications, chosen at priori, on the main geometric parameters that characterize the geometry of the component of interest.
Following this logic and working method, the tool builders have specifically created batch files with the aim of automating and speeding up this preparation phase.

Taking the CFX tool as a particular example, it is possible to create two files, listed below:

CCL file: it contains the setting created in the current state at the moment in which this file is saved. Therefore, in the initial preparation phase of the model, it is possible to 1) import the new meshes 2) move, rotate, mirror for a correct position of the component 3) load the previously saved ccl 4) Assign mesh locations to the domains and boundary conditions; in this way is possible to obtain the same model but with different geometries and meshes.

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CSE file: by enabling the "recording session" option at the start the model preparation phase, it is possible to record all the instructions executed during this phase, therefore including also the previous steps 1) 2) and 4). At the end of recording session is possible to save a file called .cse. Therefore, using this batch allows you to automate the generation of .def file completely overcoming the preparation phase of the model.

For further investigation on the topic use [4] as reference.

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7. Acceptance criteria.

This procedure is applicable for any geometry of impeller, suction, cross-over, volute and diffuser (Table 5, Table 6).

In the following tables are described the different typologies of rotor and static component which constitute a centrifugal pump:

Type impeller

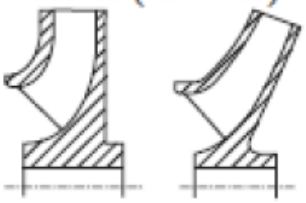
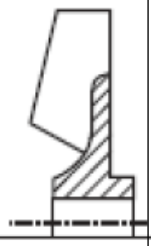
Component or feature	Radial	Semi-axial	Axial
(A) Impeller form: Characteristic: flow direction at impeller exit			
(B) Impeller type	Closed (shrouded) 	Semi-open	Open 

Table 1 : Type of geometry impeller [1]

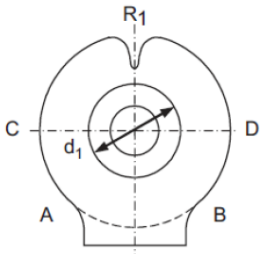
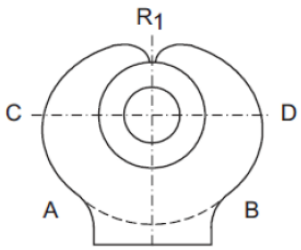
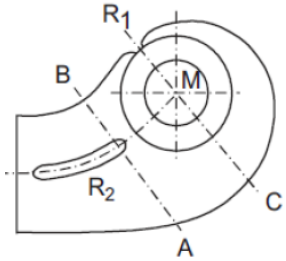
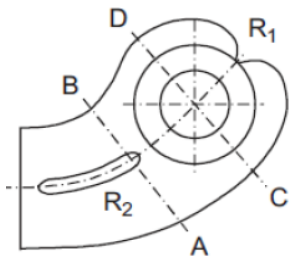
Type suction casings			
Barrel pump (radial split)		Double suction (axial split)	
anulus	Spiral shaped	asymmetrical	quasi-symmetrical
			

Table 2 : Type of geometry suction [1]

By the type of pump will be necessary to work on different composition of this part of pump.

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6. References

- *[1] Gulich "Centrifugal Pumps" ,4th.ed.*
- *[2] Ansys User Manual*
- *[3] https://www.cfd-online.com/*
- *[4]* SOP 517 "Running CFD in batch mode"

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Annex A: Pump type (*API STANDARD 610)*

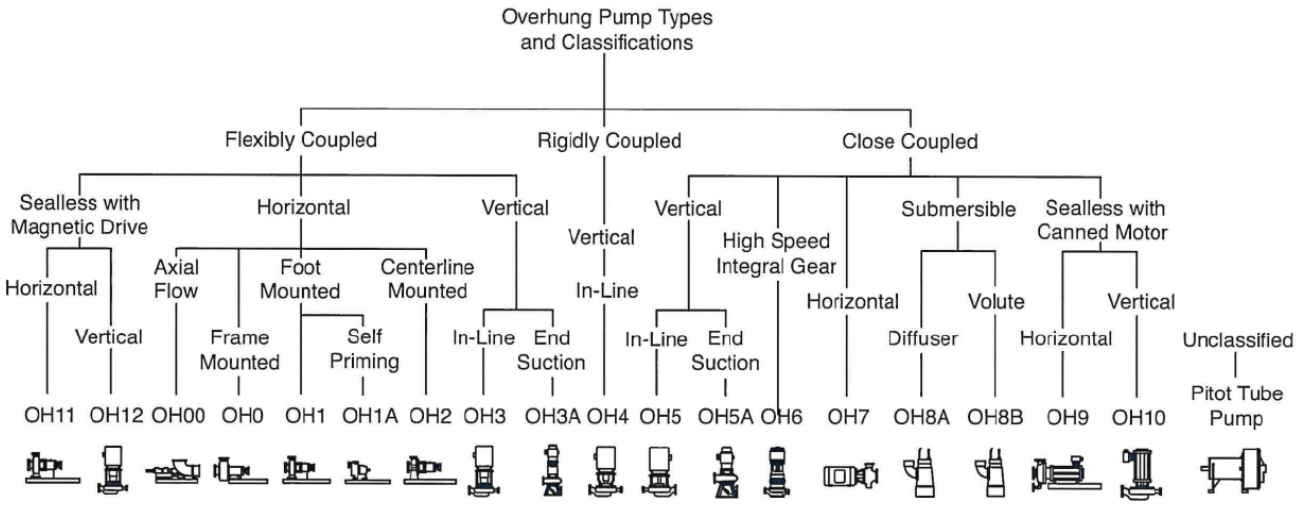

Figure 1 : Overhung pumps

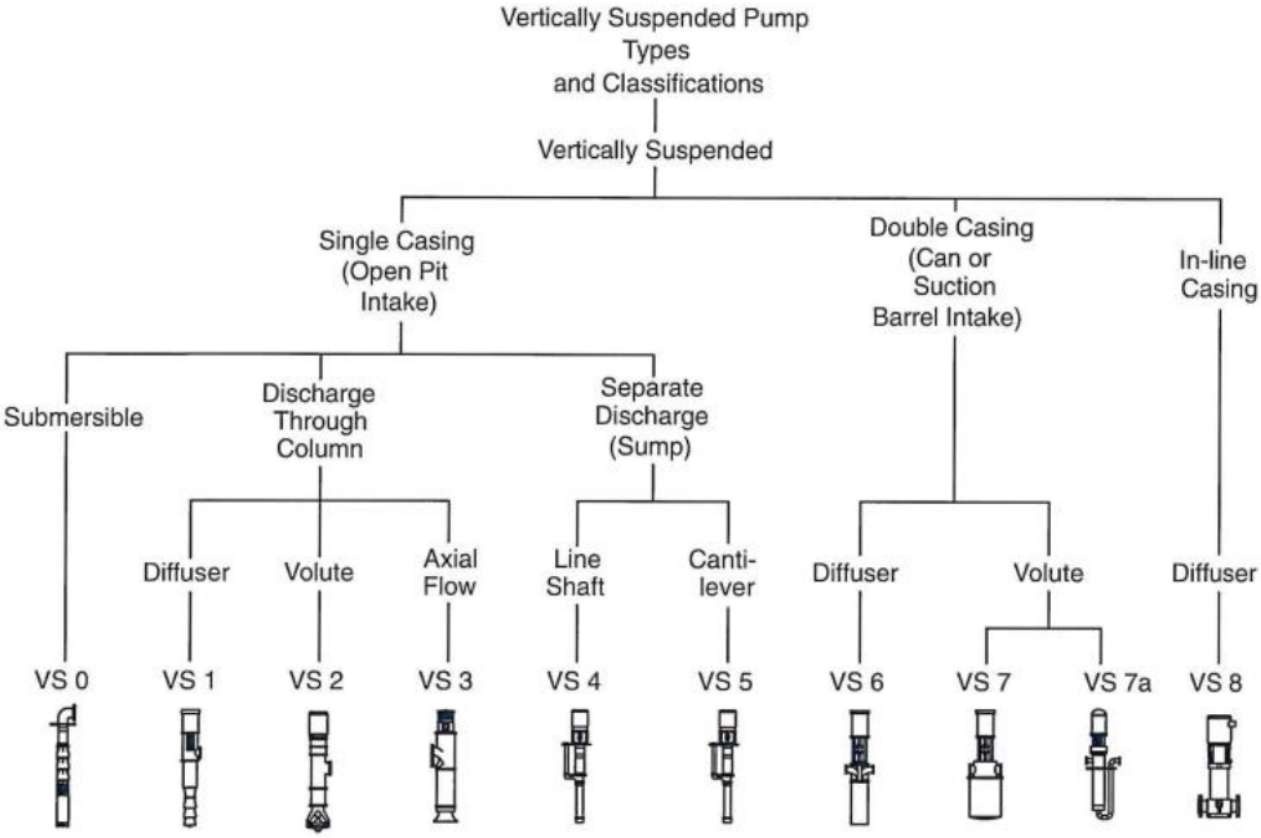

Figure 2 : Vertically suspended pumps

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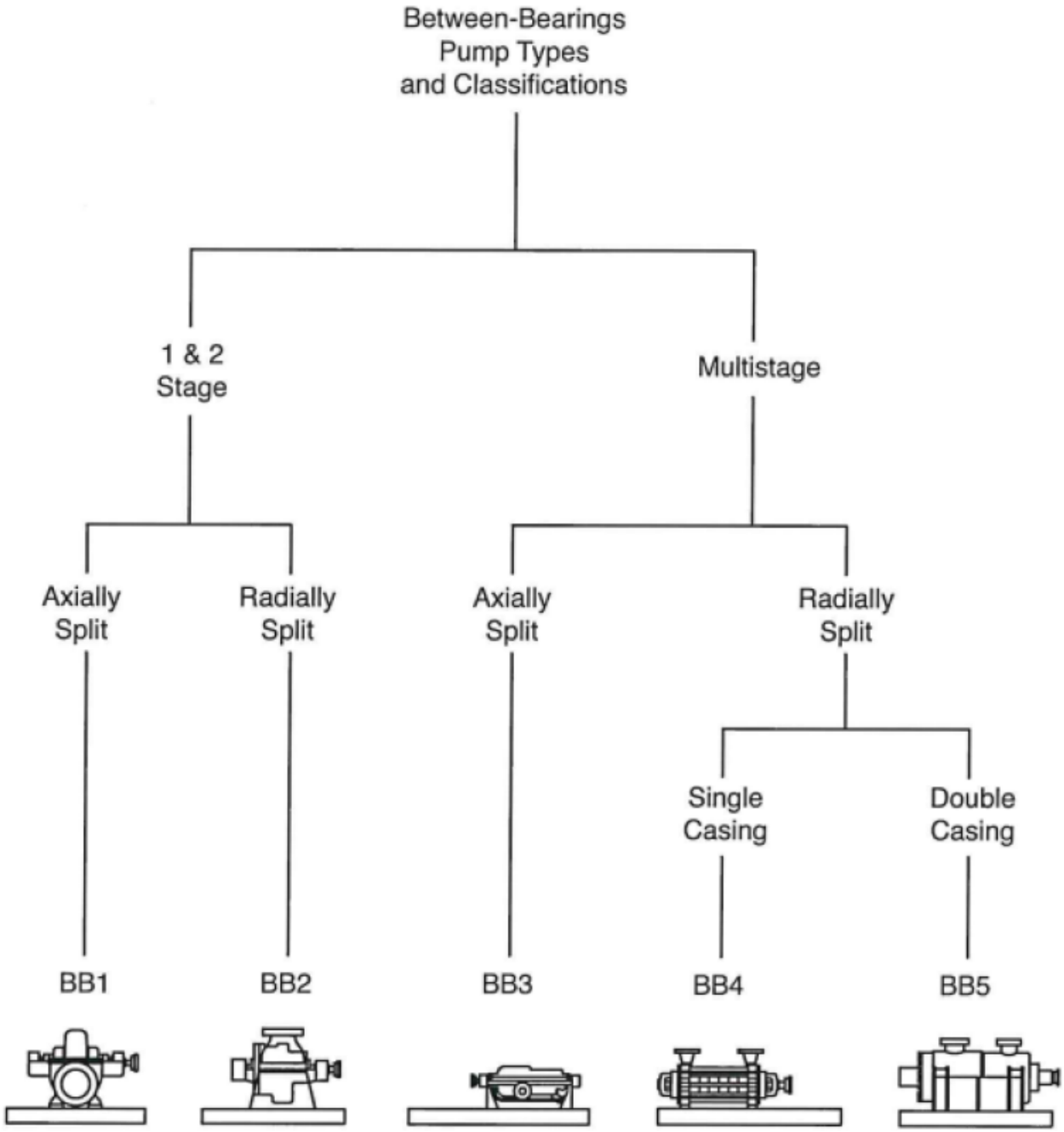

Figure 3 : Between bearings pumps

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